

Notes: Khandelwal (RES, 2010)
The Long and Short (of) Quality Ladders

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1 Big picture

- **Question.** How can we measure export quality when product quality is unobserved, and does the scope for quality differentiation affect how low-wage import competition changes U.S. manufacturing employment and output?
- **Core idea.** Prices alone are imperfect proxies for quality because prices also reflect costs, horizontal differentiation, tariffs, and freight. The paper instead infers quality from both prices and quantities: conditional on price, a product-country variety with a larger market share is assigned higher quality.
- **Setting.** The empirical setting is U.S. import data from 1989 to 2001, using manufacturing products at the ten-digit HS level nested inside SITC industries.
- **Measurement contribution.** The paper estimates quality from a nested logit demand system. Quality is the common valuation attached by U.S. consumers to an imported variety after accounting for price, nest substitution, exporter size, and time effects.
- **Quality ladders.** A product's quality ladder is measured as the range, or related dispersion measure, of estimated qualities across exporters within that product. Long ladders imply large scope for vertical differentiation; short ladders imply limited scope for quality upgrading.
- **Main empirical fact.** Richer and more capital-abundant countries export higher quality products. But the length of the quality ladder differs substantially across products and industries.
- **Labor-market result.** Low-wage import penetration reduces U.S. employment and output more strongly in industries with short quality ladders. Long-ladder industries are more insulated because U.S. firms can specialize in higher quality segments.
- **Economic takeaway.** Exposure to globalization depends not only on factor intensity or productivity, but also on whether an industry has enough vertical differentiation for high-wage producers to escape low-wage competition through quality upgrading.

2 Incremental contribution of the paper

- **From prices to demand-based quality.** The paper moves beyond the common assumption that unit values are direct proxies for quality. It uses both price and quantity information to recover a demand-based quality measure.
- **A simple and scalable quality estimator.** The nested logit approach can be implemented with standard disaggregated trade data and does not require special product-characteristic data.
- **Product-level quality ladders.** The paper introduces quality-ladder length as an empirical measure of the scope for vertical differentiation within a product market.
- **A bridge between trade quality and labor outcomes.** The paper connects quality specialization to the employment effects of low-wage import competition, showing that the same import shock can have different effects depending on the quality ladder.
- **An explanation for heterogeneous industry decline.** Traditional variables such as capital intensity, skill intensity, or TFP do not fully explain why some U.S. industries are more vulnerable to import competition. Quality-ladder length adds a distinct channel.

- **A correction to the unit-value view.** The paper shows that unit values track estimated quality more closely in long-ladder markets but are poor quality proxies in short-ladder markets, where horizontal differentiation can support high prices despite low average quality.
- **Policy relevance.** The findings help identify “contestable jobs”: jobs in short-ladder industries are more exposed to low-wage competition because firms have less room to escape by moving up the quality ladder.

3 Data and measurement

3.1 Import data

- The paper uses U.S. product-level import data from 1989 to 2001.
- Varieties are country-product pairs: a country exporting a ten-digit HS product to the United States.
- The data include import values, quantities, duties, and transportation costs.
- Unit value is defined as value plus duties plus transportation costs divided by quantity, then deflated to real terms.
- The sample is restricted to manufacturing products and excludes Rauch homogeneous goods because those products are less likely to exhibit meaningful quality differentiation.

Interpretation: The key empirical strength is the very fine product-level information. This allows the paper to infer quality within narrowly defined products, rather than only at broad industry or country levels.

3.2 Low-wage import penetration

- Low-wage countries are defined as countries with less than 5% of U.S. per capita GDP.
- Low-wage import penetration is computed at the industry-year level as total import penetration multiplied by the value share of imports from low-wage countries.
- Other import penetration is defined analogously for imports from the rest of the world.

Interpretation: The distinction between low-wage and other import penetration matters because the model predicts that low-wage competition is especially relevant for contestability of U.S. jobs.

3.3 Quality ladders

For each product h , the baseline product ladder is conceptually

$$Ladder_h = \lambda_h^{\max} - \lambda_h^{\min},$$

where $\lambda_h^{\max}$ and $\lambda_h^{\min}$ are the maximum and minimum estimated quality within product h .

The industry quality ladder is the import-share weighted average of product ladders within a four-digit SIC industry:

$$IndLadder_m = \sum_{h=1}^{H_m} w_h Ladder_h.$$

Interpretation: Long-ladder industries have more room for vertical differentiation. Short-ladder industries have less room for firms to move away from low-cost competitors by offering higher quality.

4 Model and empirical specifications

4.1 Simple model of contestable jobs

The theoretical model has firms in the North and South producing horizontally and vertically differentiated varieties. A consumer's indirect utility from variety j is

$$V_{nj} = \theta \lambda_j - \alpha p_j + \varepsilon_{nj}.$$

Quality is the vertical component. Horizontal differentiation enters through the idiosyncratic taste shock. Northern firms have higher technology and choose higher quality, while Southern firms have lower wages.

The quality ladder is proportional to

$$Ladder(\theta) = \theta \lambda_N^* - \theta \lambda_S^*.$$

Economic message: When the ladder is long, Northern firms can occupy the high-quality segment and are less directly exposed to Southern low-wage competition. When the ladder is short, quality upgrading is limited and Southern cost advantages bite harder.

4.2 Nested logit demand and quality estimation

The empirical demand equation is based on a nested logit system:

$$\ln(s_{cht}) - \ln(s_{0t}) = \lambda_{1,ch} + \lambda_{2,t} + \alpha p_{cht} + \sigma \ln(ns_{cht}) + \gamma \ln(pop_{ct}) + \lambda_{3,cht}.$$

- s_{cht} is the overall market share of country c in product h at time t .
- s_{0t} is the outside share, proxied using domestic alternatives or non-import consumption.
- p_{cht} is the c.i.f. unit value including duties and transportation costs.
- ns_{cht} is the within-product nest share.
- pop_{ct} controls for hidden varieties that may inflate observed market shares.

Estimated quality is defined as

$$\lambda_{cht} = \hat{\lambda}_{1,ch} + \hat{\lambda}_{2,t} + \hat{\lambda}_{3,cht}.$$

Interpretation: Conditional on price and market size, a variety with a larger market share receives a higher quality estimate.

4.3 Employment and output regressions

The main industry-level regression is

$$\ln Emp_{mt} = \alpha_m + \alpha_t + \beta_1 OTHPEN_{mt} + \beta_2 LWPEN_{mt} + \beta_3 (LWPEN_{mt} \times \ln IndLadder_m) + \nu_{mt}.$$

The same specification is also estimated with industry output as the dependent variable. The key coefficient is β_3 . The prediction is $\beta_3 > 0$: long quality ladders should weaken the negative effect of low-wage import penetration on U.S. employment and output.

5 Identification and empirical design

5.1 Identifying quality

- Quality is inferred from the demand residual after controlling for price, nest share, variety fixed effects, year fixed effects, and exporter size.
- Price is potentially endogenous because high-quality varieties may set higher prices.
- The paper instruments price using transportation costs, exchange rates, and distance interacted with oil prices.
- Nest shares are instrumented using the number of varieties within a product and the number of varieties exported by a country.

Interpretation: Identification comes from demand shifts and supply-side cost shifters that help separate high price due to quality from high price due to costs.

5.2 Identifying labor-market effects

- The quality ladder is measured using the initial period, reducing concerns that later employment changes mechanically affect ladder length.
- Industry fixed effects absorb time-invariant industry characteristics.
- Year fixed effects absorb common aggregate shocks.
- Instrumental-variable specifications address potential endogeneity of import penetration.
- Additional specifications control for capital intensity, TFP, and two-digit SIC-year shocks.

Interpretation: The key empirical comparison is between industries facing similar low-wage penetration but differing in quality-ladder length.

5.3 Main limitations

- Quality is inferred from demand, not directly observed.
- The nested logit imposes structure on substitution patterns.
- Product ladders are based on U.S. import data and may not fully capture global quality frontiers.
- The labor-market regressions are industry-level associations, not plant-level causal estimates.

6 Tables and figures

6.1 Table 1 and Table 2: Low-wage countries and summary statistics

Read:

- Table 1 defines the low-wage country group used to construct low-wage import penetration.
- Table 2 summarizes the product coverage, average exporter income, factor intensities, and quality-ladder lengths by broad SIC sector.
- Apparel and leather imports come from relatively low-income exporters, while transportation, industrial machinery, and chemicals are dominated by richer exporters.

Economic message: The definition of low-wage competition and the cross-sector distribution of ladder lengths provide the foundation for the labor-market analysis.

TABLE 1
Low-wage countries

Afghanistan	Chad	Haiti	Niger
Albania	China	India	Pakistan
Angola	Congo	Kenya	Rwanda
Armenia	Equatorial Guinea	Lao PDR	Samoa
Azerbaijan	Ethiopia	Madagascar	Sierra Leone
Bangladesh	Gambia	Malawi	Sri Lanka
Benin	Georgia	Mali	Sudan
Burkina Faso	Ghana	Mauritania	Togo
Burundi	Guinea	Moldova	Uganda
Cambodia	Guinea-Bissau	Mozambique	Vietnam
Central African Republic	Guyana	Nepal	Yemen

Notes: The table provides the list of low-wage countries used in the paper. Low-wage countries are defined as countries with less than 5% of US *per capita* GDP.
Source: Bernard, Jensen and Schott (2006).

Table 1: Low-wage countries

TABLE 2
Summary statistics

Sector (SIC-2)	Industry (SITC-5) (1)	Products (HS-10) (2)	Average GDP (3)	Skill intensity (4)	Capital intensity (5)	Ladder length (6)
20 Food	8	37	16,881	0.39	81.4	2.04
22 Textile	85	1,642	13,304	0.15	48.7	2.62
23 Apparel	68	2,560	7,120	0.18	11.2	2.28
24 Lumber	20	262	12,634	0.20	36.3	1.81
25 Furniture	5	72	11,849	0.25	22.1	1.66
26 Paper	38	216	19,766	0.30	126.0	1.88
27 Printing	16	55	17,574	0.87	33.2	1.33
28 Chemicals	231	2,558	20,094	0.75	166.1	2.46
29 Petroleum	7	21	10,952	0.51	509.1	2.64
30 Rubber and plastic	45	515	14,119	0.29	48.5	2.32
31 Leather	17	403	6088	0.19	18.6	1.68
32 Stone and ceramic	57	357	15,133	0.29	78.6	2.15
33 Primary metal	98	1,372	16,864	0.29	157.1	2.21
34 Fabricated metal	78	599	17,364	0.35	53.0	1.42
35 Industrial machinery	169	1,632	21,035	0.57	63.1	2.40
36 Electronic	100	1,325	15,551	0.56	57.7	2.01
37 Transportation	43	372	23,096	0.52	68.6	2.08
38 Instruments	60	715	21,843	0.96	45.3	2.40
39 Miscellaneous	76	375	10,804	0.38	29.7	2.17

Notes: The table provides summary statistics for the two-digit SIC (1987 revision) sectors. Column 1 reports the number of SITC (revision 2) industries. Column 2 reports the total number of HS products. Column 3 reports the weighted average of exporter *per capita* GDP. Columns 4 and 5 report skill (ratio of production to non-production workers) and capital intensity. Column 6 reports the (log) weighted average industry ladder [see equation (20)].
Source: NBER Manufacturing Database, Feenstra, Grossman and Helpman (2002), World Development Indicators and author's calculation.

Table 2: Summary statistics

6.2 Table 3 and Table 4: Quality estimation and factor endowments

Read:

- Table 3 summarizes the nested logit estimation across 1059 manufacturing industries.
- The IV price coefficient is more negative than the OLS coefficient, consistent with price endogeneity.
- Table 4 shows that estimated quality is positively related to exporter GDP per capita and capital abundance.
- The table also shows why controlling for hidden varieties matters: without that correction, China's large market shares can mechanically raise its apparent quality.

Economic message: The quality estimates behave plausibly: richer and more capital-abundant countries tend to export higher-quality products within narrowly defined product categories.

TABLE 3
Quality estimation results

Statistic	Mean	Median	First quarter	Third quartile
OLS price coefficient	-0.053	-0.003	-0.023	-0.0002
IV price coefficient	-0.089	-0.009	-0.070	-0.0003
*Own price elasticity	-1.28	-0.58	-1.44	-0.20
Coefficient on conditional market share	0.42	0.46	0.09	0.81
Coefficient on population	1.17	0.68	-1.04	3.81
<i>T</i> -statistic of quality estimates	10.33	3.90	1.34	10.66
Over identifying restrictions <i>p</i> -value	0.234	0.116	0.015	0.371
First stage <i>F</i> -stat <i>p</i> -value, price	0.029	0.000	0.000	0.004
First stage <i>F</i> -stat <i>p</i> -value, nest share	0.031	0.000	0.000	0.006
<i>R</i> -squared	0.24	0.14	0.04	0.40
Observations per estimation	1176	509	223	1168
Estimations with stat. sig. price coefficient			0.71	
Observations with stat. sig. price coefficient			0.81	
Total estimations			1059	
Total observations across all estimations			1,245,006	

Notes: The two-stage least squares estimation statistics of random equation (15) presented for each of the 1059 manufacturing industries.

Table 3: Quality estimation results

TABLE 4
Quality and factor endowments

Regressors	Quality _{est}		Unadjusted quality _{est}	
	(1)	(2)	(4)	(5)
Log (PCGDP _{it})	0.801***		0.025	0.085**
	0.221		0.054	0.036
Log (K/L _{it})		0.804***		
		0.210		
Log (education _{it})			0.274	
			0.207	
China dummy				0.783***
Product × year FEs	Yes	Yes	Yes	Yes
<i>R</i> -squared	0.20	0.21	0.32	0.49
Observations	1,244,475	1,172,510	512,329	1,244,475

Notes: Table regresses the quality estimates on (log) *per capita* GDP, (log) capital-labour ratios and percentage of workforce with tertiary education. Regressions include product-year fixed effects. Robust standard errors are clustered by exporting country. Column 4 uses quality estimates that do not control for hidden varieties [qualities obtained from estimating equation (14)]. Column 5 also uses the unadjusted quality measures and includes a China indicator variable. Regressions include product-year fixed effects. Robust standard errors are clustered by exporting country. Significance levels: ***0.01; **0.05; *0.10.

Table 4: Quality and factor endowments

6.3 Table 5 and Figure 1: Prices, market shares, and quality

Read:

- Table 5 shows that the relationship between prices and estimated quality is stronger in long-ladder products.
- Figure 1 contrasts two products. For transmission receivers, prices and quality move more closely together. For leather footwear, expensive varieties can have low estimated quality when they have small market shares conditional on price.
- This illustrates why price alone can be a poor quality proxy in short-ladder markets.

Economic message: The paper does not reject unit values everywhere; rather, it shows that unit values work better as quality proxies when vertical differentiation is strong.

TABLE 5 Relationship between quality and price		
Regressors	ln(price _{cht})	
	(1)	(2)
Quality _{cht}	-0.021*	-0.025*
	0.012	0.014
	0.005**	0.006**
Quality _{cht} × log (Ladder _{it})	0.002	0.003
		9.40E-05
Quality _{cht} × BW Sigma _{it}		7.20E-05
Product × year FEs	Yes	Yes
R-squared	0.87	0.86
Observations	1,245,006	1,165,703

Notes: Column 1 regresses variety quality on (log) price and its interaction with the product's baseline quality ladder. Column 2 includes quality interacted with the elasticity of substitution measures obtained from Broda and Weinstein (2004). Products with elasticities above the 90th percentile (elasticities greater than 76.9) are excluded. Robust standard errors are clustered by exporting country. Significance levels: ***0.01; **0.05; *0.10.

Table 5: Relationship between quality and price

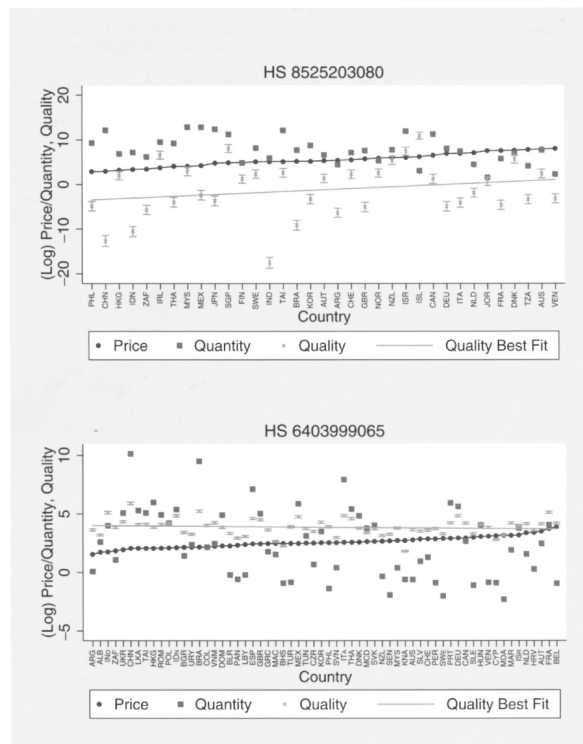


FIGURE 1
Prices, market shares, and quality. The graph plots countries' export price, quantity, and quality (with 95% confidence interval) in HS 8525203080 (transmission receivers exceeding 400 MHz) and HS 6403999065 (footwear with plastic soles, leather uppers) in 2001. Countries are ordered by unit value

Figure 1: Prices, quantities, and quality

6.4 Table 6 and Table 7: Industry characteristics and employment regressions

Read:

- Table 6 relates industry quality ladders to standard industry characteristics such as capital intensity, skill intensity, TFP, marketing, and R&D.
- Long-ladder industries are positively related to capital intensity and R&D, but standard observables explain only a small portion of ladder variation.
- Table 7 shows that low-wage import penetration is associated with lower employment, while the interaction with the quality ladder is positive.
- The positive interaction means that long-ladder industries lose less employment for the same low-wage penetration shock.

Economic message: Quality-ladder length captures a distinct dimension of industry exposure to globalization beyond capital intensity and TFP.

TABLE 6
Quality ladder and industry characteristics

Regressors	Log (IndLadder _{it})			
	(1)	(2)	(3)	(4)
Log (K/L _{it})	0.133**	0.124**		0.138**
	0.051	0.057		0.060
Log (Skill _{it})	-0.015	-0.105		-0.121
	0.067	0.080		0.080
Log (TFP _{it})	1.052*	0.934		0.833
	0.600	0.577		0.580
(Marketing intensity) _{it}		0.052		0.238
		0.866		0.853
(R&D intensity) _{it}		6.036**		4.295
		2.708		2.662
(Broda-Weinstein elasticity) _{it}			0.016	0.015
			0.011	0.011
(Coefficient of variation of unit values) _{it}			0.362	0.366
			0.230	0.258
R-squared	0.03	0.05	0.02	0.05
Observations	325	306	325	306

Notes: Column 1 regresses the industry quality ladder [defined in equation (20)] on four-digit SIC factor intensities: capital intensity, skill intensity, and total factor productivity. Capital intensity is the ratio of capital stock to total employment. Skill intensity is measured as the ratio of non-production to production workers. Total factor productivity is obtained from a five-factor model (Bartelsman, Becker and Gray, 1996). Factor intensities measured at 1989 values. Column 3 includes marketing expense to sales and R&D to sales ratios obtained from the 1975 FTC Line of Business survey. Column 3 includes industry-level elasticity of substitution (obtained from Broda and Weinstein, 2006) and industry-level coefficient of variation of unit values; these two product-level variables are aggregated to the

Table 6: Quality ladder and industry characteristics

TABLE 7
Employment regressions

Regressors	Log (employment _{it})							
	OLS				IV			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
OTHPEN _{it}	-0.493***	-1.175***	-0.475***	-0.495***	-1.774	-1.094	-0.979**	-0.745
	0.157	0.406	0.132	0.157	1.295	2.869	0.436	0.545
LWPEN _{it}	-1.894***	-2.190***	-4.643***	-1.897***	-7.162**	-8.495	-11.713***	-5.429***
	0.256	0.360	1.194	0.267	2.983	5.511	3.010	1.753
Log (IndLadder _{it}) × LWPEN _{it}	0.609**	0.716***	0.600**	0.617**	4.620*	5.714	2.879***	3.033**
	0.216	0.234	0.276	0.239	2.414	4.453	0.885	1.462
Log (IndLadder _{it}) × OTHPEN _{it}		0.136				-0.535		
						1.772		
Log (K/L _{it}) × LWPEN _{it}			0.918**				2.223**	
			0.361				0.932	
Log (TFP _{it}) × LWPEN _{it}				0.329				7.003*
				0.870				3.726
Over identification	-	-	-	-	0.48	0.43	0.38	0.40
p-value								
Kleibergen-Paap	-	-	-	-	0.61	0.61	18.40	3.52
F-statistic								
Industry-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.112	0.120	0.124	0.112	0.013	0.009	0.018	0.020
Observations	2585	2585	2585	2585	2585	2585	2585	2585

Notes: The dependent variable for each regression is the four-digit SIC industry (log) employment. The first column regresses employment on import penetration from the rest of the world (OTHPEN), low-wage import penetration (LWPEN), and the interaction of LWPEN with the industry quality ladder. Column 2 includes the OTHPEN-ladder interaction. Column 3 includes the interaction of LWPEN with initial industry capital intensity (in 1989) and column 4 includes the interaction of LWPEN with initial industry TFP (in 1989). Columns 5–8 present the IV results. The instruments are weighted average tariff rates, exchange rates, and freight rates for low-wage countries and the rest of the world. Robust standard errors are clustered at the two-digit SIC. Significance levels: ***0.01; **0.05; *0.10.

Table 7: Employment regressions

6.5 Table 8 and Table 9: Output regressions and sector-year fixed effects

Read:

- Table 8 repeats the employment analysis using output as the dependent variable.
- The output results show that the employment patterns are not simply driven by substitution of capital for labor.
- Table 9 adds two-digit SIC by year fixed effects, using only within-sector variation.
- The ladder interaction remains positive, showing that the result is not driven only by broad sector-level shocks.

Economic message: Low-wage competition reduces both employment and output more in short-ladder industries, and the result survives demanding sector-year controls.

TABLE 8
Output regressions

Regressors	Log (Output _{it})							
	OLS				IV			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
OTHPEN _{it}	-0.508**	-1.050**	-0.485**	-0.487**	-0.430	1.159	0.231	0.321
	0.200	0.497	0.181	0.190	1.980	3.834	0.284	0.857
LWPEN _{it}	-2.008***	-2.243***	-5.483**	-1.978***	-7.941**	-11.054	-16.195***	-6.619**
	0.354	0.404	1.942	0.281	4.016	7.713	3.593	2.603
Log (IndLadder _{it}) × LWPEN _{it}	0.699**	0.783**	0.687*	0.634**	5.687*	8.241	3.671***	4.439**
	0.307	0.308	0.367	0.301	3.227	6.300	1.371	1.971
Log (IndLadder _{it}) × OTHPEN _{it}		0.252				-1.248		
		0.186				2.178		
Log (K/L _{it}) × LWPEN _{it}			1.160				3.562***	
			0.749				1.379	
Log (TFP _{it}) × LWPEN _{it}				-2.589***				8.813*
				0.800				4.973
Over-identification <i>p</i> -value	—	—	—	—	0.21	0.18	0.44	0.36
Kleibergen–Paap <i>F</i> -statistic	—	—	—	—	0.61	0.61	18.40	3.52
Industry-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>R</i> -squared	0.181	0.185	0.195	0.184	0.006	0.003	0.016	0.015
Observations	2585	2585	2585	2585	2585	2585	2585	2585

Notes: The dependent variable for each regression is the four-digit SIC industry (log) output. The first column regresses output on import penetration from the rest of the world (OTHPEN), low-wage import penetration (LWPEN), and the interaction of LWPEN with the industry quality ladder. Column 2 includes the OTHPEN–ladder interaction. Column 3 includes the interaction of LWPEN with initial industry capital intensity (in 1989) and column 4 includes the interaction of LWPEN with initial industry TFP (in 1989). Columns 5–8 present the IV results. The instruments are weighted average tariff rates, exchange rates, and freight rates for low-wage countries and the rest of the world. Robust standard errors are clustered at the two-digit SIC. Significance levels: ***0.01; **0.05; *0.10.

Table 8: Output regressions

6.6 Table 10: Robustness checks

Read:

- Table 10 reports robustness checks using alternative ladder definitions and alternative quality-estimation choices.
- Results remain positive and significant when quality is estimated with tariffs as extra instruments, when quality excludes residuals, when GDP replaces population as the hidden-variety proxy, and when the ladder is measured using range, interquartile range, or standard deviation.
- Count-based measures and Broda-Weinstein horizontal-substitution measures do not replicate the baseline pattern as cleanly.

Economic message: The key force is vertical differentiation rather than simple product counts or horizontal substitutability.

TABLE 9
Employment and output regressions, SIC₂-year fixed effects: IV

Regressors	IV–Log (Employment _{it})			IV–Log (Output _{it})		
	(1)	(2)	(3)	(4)	(5)	(6)
OTHPEN _{it}	-0.720	-0.512	-0.499	-0.066	0.209	0.188
	1.234	0.496	0.675	1.529	0.345	0.699
LWPEN _{it}	-4.398***	-10.558**	-4.531***	-4.486***	-13.000***	-4.471**
	1.344	4.625	1.680	1.307	2.124	1.874
Log (IndLadder _{it}) × LWPEN _{it}	2.706**	2.148**	2.764**	3.275**	2.082**	3.045**
	1.360	0.954	1.357	1.590	1.059	1.204
Log (K/L _{it}) × LWPEN _{it}		2.121			2.946***	
		1.440			0.890	
Log (TFP _{it}) × LWPEN _{it}			6.777			6.452
			4.717			4.583
Over-identification <i>p</i> -value	0.36	0.35	0.59	0.281	0.339	0.435
Kleibergen–Paap <i>F</i> -statistic	1.13	2.91	36.40	1.13	2.91	36.4
Industry-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Two-digit SIC × year FEs	Yes	Yes	Yes	Yes	Yes	Yes
<i>R</i> -squared	0.008	0.018	0.005	0.006	0.000	0.007
Obs.	2585	2585	2585	2585	2585	2585

Notes: The dependent variables for each regression in columns 1–3 and columns 4–6 are the four-digit SIC industry (log) employment and output, respectively. Each regression includes two-digit SIC–year pair fixed effects. The first column regresses employment on import penetration from the rest of the world (OTHPEN), low-wage import penetration (LWPEN), and the interaction of LWPEN with the industry quality ladder. Column 2 includes the interaction of LWPEN with initial industry capital intensity (in 1989) and column 3 includes the interaction of LWPEN with initial industry TFP (in 1989). Columns 4–6 present the IV results. The instruments are weighted average tariff rates, exchange rates, and freight rates for low-wage countries and the rest of the world. Robust standard errors are clustered at the two-digit SIC. Significance levels: ***0.01; **0.05; *0.10.

Table 9: Employment and output, SIC2-year fixed effects

TABLE 10
Employment regressions, robustness checks

LWPEN–ladder interaction	Regression method	
	OLS	IV
(1) Quality identified from baseline instruments and tariffs	0.501*	2.755**
	0.271	1.328
(2) Product count	0.000	−0.005
	0.001	0.005
(3) Variety count	−0.440	0.690
	0.348	0.948
(4) Price	1.285	7.436*
	1.227	4.129
(5) Market share	12.664***	21.512
	3.415	21.914
(6) Quality excluding residuals	0.646***	4.571*
	0.217	2.435
(7) GDP–quality ladder	0.992**	5.563***
	0.348	1.986
(8) Broda–Weinstein ladder	−0.910*	−1.591
	0.471	1.141
(9) Exclude MFA products	0.484**	2.711**
	0.232	1.378
(10) Range	0.639**	3.888***
	0.233	1.339
(11) Inter-quartile range	0.650**	5.236**
	0.239	2.667
(12) Standard deviation	0.751**	5.061**
	0.271	2.417

Notes: Table re-runs the employment regression in equation (23) and reports the industry ladder interaction with LWPEN coefficient for alternative definitions of the quality ladder. Row 1 re-estimates equation (15) to include tariffs as additional instruments. Row 2 constructs the ladder from the number of countries (i.e. varieties) within the product and then aggregates to the industry according to equation (20). Row 3 counts the number of products within the four-digit industry code. Rows 4 and 5 construct the ladder from the dispersion of prices and market shares within products, respectively. Row 6 defines the ladder from quality estimates exclusive of the residual: $\lambda_{chr} = \lambda_{1,chr} + \lambda_{2,j}$. Row 7 re-runs equation (15) but uses GDP instead of population to control for hidden varieties. Row 8 constructs the quality ladder from the HS-level elasticities of substitution estimated by Broda and Weinstein (2006). Row 9 excludes HS codes that were subject to quotas under the MFA in the construction of the industry ladders; the HS codes that were subject to quotas are taken from Brambilla, Khandelwal and Schott (2008). The remaining rows use the full-range, inter-quartile range, and standard deviation of qualities within products. Robust standard errors are clustered at the two-digit SIC for each regression. Significance levels: ***0.01; **0.05; *0.10.

Table 10: Employment regressions, robustness checks

7 Short summary

- The paper develops a demand-based measure of export quality using prices and market shares from U.S. import data.
- Quality is inferred from a nested logit demand system: conditional on price, larger market share implies higher quality.
- The paper constructs quality ladders as the dispersion of estimated qualities within product markets.
- Richer and more capital-abundant countries tend to export higher-quality products.
- Unit values are better quality proxies in long-ladder markets, but can be misleading in short-ladder markets.
- Low-wage import penetration reduces U.S. manufacturing employment and output.
- The negative effect is substantially weaker in long-ladder industries.
- The result remains after controlling for capital intensity, TFP, broad sector-year shocks, and alternative ladder definitions.
- The main conclusion is that quality specialization helps explain which jobs and industries are more contestable by low-wage competition.

8 Conclusion

8.1 Core conclusion

Khandelwal shows that product quality can be measured more carefully by combining prices with market shares. The estimated qualities reveal that product markets differ substantially in the length of their quality ladders. This heterogeneity matters for trade exposure: industries with short quality ladders suffer larger employment and output losses when low-wage import penetration rises.

8.2 Economic intuition

The mechanism is simple. High-wage producers can survive low-wage competition if they can move into higher-quality segments. This is feasible in long-ladder markets because consumers place meaningful value on quality differences. In short-ladder markets, however, products have limited scope for vertical differentiation. U.S. firms then compete more directly with low-cost exporters, making employment and output more vulnerable.

8.3 Incremental contribution revisited

The paper contributes both a measurement tool and an economic mechanism. The measurement tool is a nested-logit estimator of quality that improves on unit values by using market shares. The economic mechanism is that the quality ladder mediates the labor-market effect of trade. This moves the literature beyond asking whether import competition hurts manufacturing toward asking which markets are structurally more exposed.

8.4 Implications for empirical work

For empirical trade research, the paper warns against treating unit values as universal quality proxies. Prices can reflect quality in some markets, but in short-ladder markets they may reflect horizontal differentiation or costs. For labor-market research, the paper suggests that industry vulnerability to globalization depends on vertical differentiation, not only on factor intensity or productivity.

8.5 Limitations

The quality estimates depend on the nested logit structure, the choice of outside good, and the validity of instruments for price and nest shares. The employment regressions are industry-level and therefore cannot fully isolate plant-level mechanisms. The quality ladders are measured from U.S. import data, so they may not capture all global quality possibilities.

8.6 Takeaway

The enduring takeaway is that globalization does not affect all industries equally. Industries with long quality ladders have a route of escape through quality upgrading; industries with short ladders are much more exposed to low-wage competition. The paper turns the abstract idea of “quality specialization” into an empirical tool for understanding trade-induced employment and output changes.